Priynka Roy

MSc Statistics Student | Data Science & Machine Learning
Industrial Analytics

Turku, Finland | priynka470@gmail.com
GitHub | LinkedIn | Google Scholar

PROFILE

Master's student in Mathematics and Statistics at the University of Turku, specializing in Statistics. A background in statistical research and independent Python projects supports my interests in time-series analysis, machine learning, and industrial decision support. Seeking opportunities in data science, statistical analytics, and energy or equipment analytics.

SELECTED INDEPENDENT PROJECTS

IndustrialGuard AI

Explainable equipment condition monitoring | Python, scikit-learn, Streamlit

- Built an eight-section application using real MetroPT-3 compressor telemetry, with causal time-series features, robust statistics, regularized covariance monitoring, and Isolation Forest.
- Compared frozen monitoring policies chronologically; exposed alert burden, operating-regime effects, and feature deviations. Sparse historical failures limit conclusions about prediction.

Source code & methodology | Live application

EnerTwin AI

Industrial energy forecasting & constrained scheduling | Python, SciPy HiGHS, Streamlit

- Developed five 24-hour forecasting methods on 35,040 electricity observations. Gradient boosting achieved MAE 10.870 and RMSE 19.408 kWh per interval on the original October-December holdout.
- Implemented forecast uncertainty diagnostics, historical residual scenarios, and deterministic and robust linear-programming schedules with feasibility checks and frozen-decision backtests.
- Documented calibration transfer limits and hypothetical tariffs and workloads; modeled cost comparisons do not establish realized industrial savings.

Source code & methodology | Live application

TECHNICAL SKILLS

Programming & data: Python, R, pandas, NumPy, SPSS
Statistics & ML: statistical inference, regression, time-series analysis, forecast evaluation, scikit-learn, anomaly detection, explainability
Optimization: linear programming, SciPy optimization, scenario analysis, robust scheduling
Tools & visualization: Power BI, Microsoft Excel, Plotly, Streamlit, Git, GitHub

Priynka Roy

Education, research & recognition

EDUCATION

Master of Science in Mathematics and Statistics

University of Turku | Turku, Finland | August 2025 - Present

Specialization: Statistics. Current academic interests include statistical modeling, time-series analysis, and statistical computing.

Bachelor of Science in Statistics

Faculty of Science, Mawlana Bhashani Science and Technology University
Tangail, Bangladesh | January 2016 - December 2020

RESEARCH & PUBLICATION

Peer-reviewed publication

Roy, P., Chowdhury, M. H., & Ahmed, A. (2024). Exploring determinants of basic childhood vaccination coverage in Bangladesh: A generalized Poisson regression analysis. *Discover Public Health, 21*, 179.

DOI: 10.1186/s12982-024-00311-w

Undergraduate research

Factors Associated with Stunting Among Under Five Years Children in Rangpur Division, Bangladesh. Department of Statistics, MBSTU.

Current master's research interests

Ongoing research involving statistical disclosure control and longitudinal data.

SCHOLARSHIPS

Research Scholarship
Mawlana Bhashani Science and Technology University, Tangail, Bangladesh.

Merit Scholarship
Faculty of Exact Science, Turku University, Turku, Finland.

PROFESSIONAL DEVELOPMENT

Data Analyst in Power BI career track | DataCamp
July 2026 - Present. In progress.

Workshop / conference participation
The 1st National Conference on Application of Statistics on Sustainable Development Goals, MBSTU, Bangladesh.

LANGUAGES

Bengali: native | English: IELTS 6.5 | Finnish (Suomi): A1